

Course Type	Course Code	Name of Course	L	T	P	Credit
DSC 3	NMEC103	Thermodynamics	3	0	0	3

Course Objective

The course will give a brief overview of the basic concepts of Thermodynamics followed by applications of the Laws of Thermodynamics to different systems and estimate their performance and various energy interactions for understanding the different Thermodynamic cycles. The course will also focus on Air standard cycles, Steam power cycles, gas power cycles, and Refrigeration cycles.

Learning Outcomes

Upon successful completion of this course, students will:

- understand the basics and laws of Thermodynamics
- learn the different types of power and refrigeration cycles employing gas or vapour as the working fluid
- be able to apply 1st and 2nd laws to any system to estimate their performance, determine various energy interactions and associated losses due to system irreversibility

Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
1	Basic Thermodynamics: Definition of Thermodynamics, thermodynamic system, open system, closed system and isolated system, surroundings, control volume, state point, properties, extensive and intensive properties, point and path functions (exact and inexact differentials), thermodynamic processes and cycles, thermodynamic equilibrium, Quasi-static process, concept of continuum and local thermodynamic equilibrium.	6	Understanding the basics of Thermodynamics and various definitions
2	Properties of a pure substance: Phase, Pure substance, state postulate for simple systems, p-v, p-T, T-s and h-s diagrams, significance of critical and triple point, estimation of thermodynamic properties: steam tables, ideal gas equation of state	5	Learning properties and change of properties of pure substances. Understand the methods used for estimating thermodynamic properties.
3	Work & Heat: Definition of work, various forms of work interaction, work interaction for a simple compressible pure substance undergoing frictionless quasi-static flow and non-flow processes, Heat interaction	4	Understanding the thermodynamic definition of work and heat interaction.
4	Laws of Thermodynamics: Zeroth's law of thermodynamics and its significance, First law of thermodynamics for a closed system and open system, Caratheodory's axiom I, Internal energy and enthalpy, General expression of first law of thermodynamics. Physical meaning of Second law, Statement	6	Understanding laws of thermodynamics and their applications to various systems undergoing

	of Second law, Clausius inequality, Carnot's theorem, External and internal irreversibility		thermodynamic cycle/process
5.	Entropy: Introduction to entropy, its statistical interpretation, Caratheodory's axiom II, Entropy balance equation for a closed system and control volume. Entropy measurement and its evaluation, Mechanism of entropy generation: Heat transfer across a finite temperature difference, flow with friction, mixing, entropy principle for a process,	5	Learning the concept of entropy and its applications in estimating thermodynamic losses and the direction of spontaneous processes.
5.	Air standard cycles: Air Standard Assumption Carnot cycle, Otto cycle, Diesel cycle, Dual cycles, Stirling and Ericsson cycles, and their comparison.	4	Learning cycles with air as the working fluid
6.	Vapour Power Cycles: Carnot cycle, Rankine cycle, Reheat and Regenerative cycles, work, power, efficiency	4	Learning cycles for steam power plant
7.	Gas power cycles: Open cycle and closed cycle gas turbine cycle, intercooling, reheating and regeneration effects.	4	Learning cycles for gas turbine power plant
8.	Refrigeration cycles: Reversed Carnot cycle, Vapor compression refrigeration cycle, Vapor absorption refrigeration cycle.	4	Learning cycles used for refrigeration
	Total lecture hours	42	

Text Books :

1. Fundamentals of Thermodynamics: Sonntag, Borgnakke and Van Wylen, Wiley, 6th edition
2. Fundamentals of Engineering Thermodynamics: Michael J Moran and Howard N. Shapiro, D. D. Boettner, M. B. Bailey, Wiley, 9th edition
3. Thermodynamics-An Engineering Approach: Y A Cengel and M A Boles, Mc-Graw Hill, 8th edition

Reference Books:

1. Fundamentals of Thermodynamics: Claus Borgnakke, Richard E. Sonntag, Wiley 10th edition
2. Heat & Thermodynamics: M. W. Zemansky & R. H. Dittman, Mc-Graw Hill 7th edition
3. Engineering Thermodynamics Work and Heat Transfer: G. Rogers and Y. Mayhew, Pearson, 4th edition
4. Thermodynamics from concept to application: A. Shavit & C. Gutfinger, CRC Press, 2nd edition

Marks Distribution:	
Mid Semester	32 Marks
End Semester	48 Marks
Assignments	10 Marks
Quizzes	10 Marks
Total	100 Marks